to a Big Crunch after all. Fortunately, that looks very unlikely, though we can't entirely rule it out.)

In any case, based on observations at the moment, it really looks a lot like dark energy is a cosmological constant: an unchanging property of spacetime that has only recently (i.e., in the last few billion years) come to dominate the evolution of the universe. At early times, when the cosmos was more compact, there just wasn't enough *space* for the cosmological constant (which is a property of empty space) to do very much, so the expansion at that time was slowing down, just as we would have expected. But about five billion years ago, matter got so diffuse due to ordinary cosmic expansion that the inherent cosmological-constant-induced stretchiness of space started to really become noticeable. We can now measure the motion of supernova explosions so far away that they went off before the expansion started to speed up, meaning that we can trace out when the universe was decelerating, and pretty much exactly when it transitioned to acceleration. Dark energy still might be some new, dynamical field. But so far, a cosmological constant fits the data perfectly.

If we follow that through to its future consequences, it's kind of ironic, actually. Because now it seems that the term Einstein used to save the universe will end up spelling its doom.

THE INFINITE COSMIC TREADMILL

A cosmological-constant-induced apocalypse is a slow and agonizing one, marked by increasing isolation, inexorable decay, and an eons-long fade into darkness. In some sense, it doesn't end *the universe* exactly, but rather ends *everything in it*, and renders it null and void.

The reason a cosmological constant dooms the universe is that once it starts, the accelerated expansion never, ever stops.

The present-day observable universe is probably bigger than you think. The "observable" part refers to the region within our *particle horizon*. We define this as being the farthest we could possibly see, given the limitations of the speed of light and the age of the universe. Since light takes time to travel, and more distant objects are, from our perspective, farther in the past, there has to be a distance corresponding to the beginning of time itself. A distance at which, if a light beam started there at the first moment, it would